

Athulith Paraselli

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Portfolio: <https://apasaselli.github.io/>

RESEARCH INTERESTS	Trustworthy Machine Learning, Mechanistic Interpretability, Multimodal Artificial Intelligence	
EDUCATION	Brown University , Providence, RI	Sep 2025 - May 2027
	Master of Science, Computer Science - Artificial Intelligence/Machine Learning	
	University of California, San Diego (UCSD) , La Jolla, CA	Sep 2021- June 2025
	Bachelor of Science, Mathematics-Computer Science, <i>Data Science Minor</i>	cGPA: 3.82 / 4.00
RESEARCH EXPERIENCE	Graduate Research Assistant , Brown University	Sep 2025 – Present
	<i>Advised by Dr. Ellie Pavlick</i>	
	- Designed and conducted experiment to understand visual demographic biases within VLMs. - Developed generalized pipeline for activation patching and logit lens within VLMs. - Created steering vectors and applied them at inference to allow for training-free bias correction.	
	Undergraduate Research Assistant , UCSD	Apr 2025 – Aug 2025
	<i>Advised by Dr. Xi Li and Dr. Zhuowen Tu</i>	
	- Developed novel multimodal adversarial training pipeline to improve robustness of VLMs by 12%. - Integrated LoRA and prompt tuning in a unified training loop, enabling flexible defense strategies.	
	Undergraduate Research Assistant , UCSD	Nov 2024 – Nov 2025
	<i>Advised by Dr. George Sugihara</i>	
	- Designed and implemented a hybrid-LSTM model in PyTorch, achieving 70% accuracy in forecasting algal blooms. - Fine-tuned Chronos, a foundation model for time-series forecasting, on ecological data to reduce weighted quantile loss (wQL) of algal bloom predictions by 32%.	
TECHNICAL EXPERIENCE	Software Engineer Intern , Qualcomm	Jun 2024 – Sep 2024
	Modem Integration Testing	
	- Created an efficient unit testing workflow through optimizing existing industry-standard methods. - Implemented functionality to automatically generate unit tests based on customer specifications.	
WORKING PAPERS	<i>Predicting Bioluminescent Algal Blooms by Conditioning Neural Forecasts on Reconstructed Dynamics</i> Athulith Paraselli , Ciro Zhang, and George Sugihara	
TEACHING EXPERIENCE	Undergraduate Instructional Assistant	
	<i>Conduct exam review sessions, design homework/exams, hold weekly office hours, grade assignments</i>	
	Principles of Data Science , UCSD	Apr 2024 - Jun 2025
	Advanced Machine Learning Methods , UCSD	Apr 2025 - Jun 2025
TECHNICAL SKILLS	Programming Languages Python, C/C++, Java, Perl, ARM Assembly, MATLAB, SQL, HTML, CSS, JavaScript, Svelte	
	Areas of Expertise Version Control(Git/Github), Machine Learning(Scikit-learn, Pytorch, Tensorflow, Keras, Hugging Face, NNSight), AWS(Amplify, API Gateway, Lambda, DynamoDB)	

SERVICES &
OUTREACH

Eta Kappa Nu (HKN) Projects Program

Nov 2024 – Jun 2025

Inaugural Project Lead

- Led a team of 4 students to develop a live data processing pipeline and website to display Bioluminescent algal bloom forecasts.
- Established connections between Labs at Scripps Institution of Oceanography and Eta Kappa Nu.
- Coordinated community outreach initiatives, presenting accessible explanations of Harmful Algal Blooms and sharing updates from ongoing research collaborations.
- Presented results at San Diego Undergrad Tech Conference.

Eta Kappa Nu (HKN) Events Team

Mar 2024 – Jun 2025

Math-CS Events Chair

- Organized and led technical workshops on fundamental ML concepts such as neural networks and gradient descent, enhancing accessibility for underclassmen.
- Collaborated with board members to coordinate events, recruit inductees, and promote technical engagement across the department.

AWARDS &
HONORS

- Eta Kappa Nu (IEEE HKN) Member, *UCSD*
- Provost Honors x11, *UCSD*

2023-2025

2021-2025